# Notes: Khorrami and Mendo (WP 2025)

## Dynamic Self-Fulfilling Fire Sales

## Contents

# 1 Big picture: what the paper does and why it matters

**Core question.** Why do fire sales and "accelerator"-type crises occur even when many fundamental risks are hedgeable? If agents can frictionlessly share (hedge) the fundamental shock that hits capital, the usual intuition is that balance-sheet amplification should disappear.

**Core answer (paper's contribution).** The paper builds a version of the Brunnermeier–Sannikov / Kiyotaki–Moore class of models in which *all fundamental risk is perfectly hedgeable*, and shows that:

(i) Fundamental shocks cannot generate fire sales in equilibrium (a safe, efficient equilibrium exists).

(ii) *Nonetheless*, there also exist equilibria with fire sales driven by *non-fundamental* (sunspot) shocks that are *not* hedgeable.

(iii) A simple refinement based on *vanishingly small perceived fundamental risk* selects the fire-sale equilibrium as the unique limiting outcome.

**Key mechanism in one sentence.** Even if markets span the fundamental shock, agents cannot hedge *endogenous* risks that arise only because they coordinate on them; those unspanned risks can generate self-fulfilling selling, price drops, and misallocation.

**Notation map.** The paper's state variable for experts' wealth share is $\eta$ in the figures, while it is denoted $h$ in the text. I use $h$ throughout (and interpret $\eta$ in figures as $h$).

**Tables vs figures.** This paper contains **no tables** (it is primarily theoretical/numerical). It contains figures in the main text and appendices. All figure pages are embedded in the appendix of these notes.

# 2 Baseline continuous-time model (Section 1)

## 2.1 Uncertainty and technology

There are two independent Brownian motions $(W_t, Z_t)$:

- $W_t$: *fundamental* shock that affects capital.
- $Z_t$: *sunspot* shock (extrinsic to primitives).

There is a durable capital good that produces the consumption good. If an agent of type $i \in \{e, h\}$ holds capital $k_{i,t}$ (abstracting from trades), its law of motion is

$$\mathrm{d}k_{i,t} = k_{i,t}\big(g \ \mathrm{d}t + \sigma \ \mathrm{d}W_t\big), \tag{1}$$

with $g > 0$ and $\sigma > 0$ constants. The market price of capital is $q_t$.

## 2.2 Agents, productivity, and markets

There are two types:

- Experts ($e$): productivity $a_e$ output per unit of capital.
- Households ($h$): productivity $a_h \in (0, a_e)$ output per unit of capital.

  Financial markets:

(a) A short-term risk-free bond in zero net supply, paying interest rate $r_t$.

(b) A market to contract on the *fundamental* Brownian shock $\mathrm{d}W_t$ with instantaneous risk price $p_t$.

**Crucial incompleteness:** there is *no* market to hedge $\mathrm{d}Z_t$.

**Financial friction.** Agents cannot issue equity nor state-contingent debt against the capital they manage. They face solvency $n_{i,t} \geq 0$ and capital short-sales constraints $k_{i,t} \geq 0$.

## 2.3 Budget constraint

Let $n_{i,t}$ denote net worth, $c_{i,t}$ consumption, $k_{i,t}$ capital holdings, and $x_{i,t}$ exposure to the $\mathrm{d}W_t$ hedge. The paper writes the net-worth SDE as

$$\mathrm{d}n_{i,t} = \underbrace{\Big[(n_{i,t} - q_t k_{i,t})r_t - c_{i,t}\Big]\,\mathrm{d}t}_{\text{bond position + consumption}} + \underbrace{q_t k_{i,t}\frac{a_i}{q_t}\,\mathrm{d}t + \mathrm{d}(q_t k_{i,t})}_{\text{capital cash flow + valuation}} + \underbrace{x_{i,t}\big(p_t\,\mathrm{d}t + \mathrm{d}W_t\big)}_{\text{fundamental hedge}}. \tag{2}$$

The exact decomposition is an accounting identity: net worth earns the bond rate on the portion not invested in capital, earns the capital dividend/production flow $a_i k_{i,t}\,\mathrm{d}t$ (scaled by $q_t$ in the paper's notation), plus capital price changes, plus hedge payoffs.

## 2.4 Preferences and optimization

Agents have log utility with discount rates $\rho_e > \rho_h > 0$:

$$\max_{\{c_{i,t},k_{i,t},x_{i,t}\}_{t\geq 0}} \mathbb{E}\int_0^\infty e^{-\rho_i t}\log(c_{i,t})\ \mathrm{d}t \quad \text{s.t. } (2),\ c_{i,t} \geq 0,\ k_{i,t} \geq 0,\ n_{i,t} \geq 0. \tag{3}$$

**Scale invariance.** Because log utility and the budget set are homogeneous of degree one, the problem is effectively mean–variance in *portfolio shares* (Appendix A provides an HJB sketch).

## 2.5 Competitive equilibrium

An equilibrium consists of processes

$$\{q_t, r_t, p_t, k_{e,t}, k_{h,t}, x_{e,t}, x_{h,t}, c_{e,t}, c_{h,t}, n_{e,t}, n_{h,t}\}_{t\geq 0}$$

such that:

(i) Given $(q_t, r_t, p_t)$, each agent solves (3).

(ii) Goods market clears:

$$c_{e,t} + c_{h,t} = a_e k_{e,t} + a_h k_{h,t}. \tag{4}$$

(iii) Capital market clears: $k_{e,t} + k_{h,t} = K_t$, where $K_t$ follows (1).

(iv) Hedge market clears: $x_{e,t} + x_{h,t} = 0$.

# 3 Equilibrium characterization (Section 1.1)

## 3.1 Capital price dynamics

The paper conjectures

$$\mathrm{d}q_t = q_t\Big(m_{q,t}\,\mathrm{d}t + \sigma_{q,t}\,\mathrm{d}W_t + \upsilon_{q,t}\,\mathrm{d}Z_t\Big), \tag{5}$$

where $\upsilon_{q,t}$ is *sunspot* price volatility. A **fundamental equilibrium** has $\upsilon_{q,t} \equiv 0$; otherwise it is a **Brownian sunspot equilibrium**.

## 3.2 Consumption rule and mean–variance portfolio choice

With log utility, optimal consumption is

$$c_{i,t} = \rho_i n_{i,t}. \tag{6}$$

Portfolio choice reduces to a mean–variance problem over exposures to $\mathrm{d}W_t$ and $\mathrm{d}Z_t$ (Appendix A):

$$\max_{k\geq 0,\, x\in\mathbb{R}} \left\{ \mathbb{E}\Big[\frac{\mathrm{d}n}{n}\Big] - \frac{1}{2}\mathrm{Var}\Big[\frac{\mathrm{d}n}{n}\Big] \right\}. \tag{7}$$

Intuition: with log utility, the HJB for $\log n$ turns the objective into mean minus half variance of the growth rate.

## 3.3 First-order conditions: capital and hedging

Define experts' and households' wealth and capital shares:

$$h_t := \frac{n_{e,t}}{n_{e,t} + n_{h,t}} = \frac{n_{e,t}}{q_t K_t}, \qquad \kappa_t := \frac{k_{e,t}}{K_t}. \tag{8}$$

Here $\kappa_t$ is the share of capital managed by experts (the paper denotes it by $k$; figures label it $\kappa$).

The portfolio optimality conditions imply:

(a) **Risk-sharing in fundamental shock $W$:** the equilibrium risk price satisfies

$$p_t = \sigma + \sigma_{q,t}. \tag{9}$$

In particular, this delivers *full insurance of fundamental risk in the wealth-share state* (see below).

(b) **Risk-balance in unhedgeable sunspot risk $Z$:** when both types are marginal in the capital market, expected-return gaps must compensate for differential exposure to $\upsilon_{q,t}\,\mathrm{d}Z_t$.

### 3.4 Aggregation: the price–output relation (PO)

Using market clearing (4) and $c_{i,t} = \rho_i n_{i,t}$, define the wealth-weighted average discount rate

$$\bar{\rho}(h) := h\rho_e + (1-h)\rho_h. \tag{10}$$

Then goods market clearing implies the key **price–output relation**:

$$q\,\bar{\rho}(h) = \kappa a_e + (1-\kappa)a_h. \tag{PO}$$

*Interpretation:* because experts value capital more (higher productivity), capital is more valuable (higher $q$) when more of it is allocated to experts (higher $\kappa$).

### 3.5 Risk-balance condition (RB) and its meaning

The paper derives

$$0 = \min\left\{1-\kappa,\ \frac{a_e - a_h}{q} - \frac{\kappa - h}{h(1-h)}\,\upsilon_q^2\right\}. \tag{RB}$$

Two regimes:

- If experts manage all capital ($\kappa = 1$), households are out of the capital market, and the second term need not bind.
- If both types hold capital ($\kappa < 1$), then the return premium from expert productivity advantage, $(a_e - a_h)/q$, equals compensation for experts' *excess exposure* to the unhedgeable sunspot shock, proportional to $(\kappa - h)\upsilon_q^2$.

### 3.6 Wealth-share dynamics and the "fundamental risk is fully shared" result

The wealth share evolves as

$$\mathrm{d}h_t = \mu_h(h_t)\,\mathrm{d}t + \sigma_h(h_t)\,\mathrm{d}W_t + \upsilon_h(h_t)\,\mathrm{d}Z_t, \tag{11}$$

and the paper shows:

$$\sigma_h(h) \equiv 0, \qquad \upsilon_h(h) = (\kappa - h)\upsilon_q. \tag{12}$$

*Interpretation:* because agents can trade a hedge on $\mathrm{d}W_t$, they can fully share fundamental risk, so the *distribution of wealth* is not hit by $\mathrm{d}W_t$. But $\mathrm{d}Z_t$ is unspanned, and experts are more exposed to capital, so a sunspot-driven drop in $q_t$ shifts wealth away from experts.

## 4 Fundamental equilibrium (FE) (Section 1.2)

### 4.1 Definition and key property

A **fundamental equilibrium** sets $\upsilon_q \equiv 0$. Then the risk-balance condition (RB) forces $\kappa \equiv 1$ (full efficiency), and the price–output relation (PO) gives

$$q(h) = \frac{a_e}{\bar{\rho}(h)}. \tag{13}$$

Because $h_0$ is pinned down by initial endowments, this equilibrium is **unique** within the Markov class.

### 4.2 Deterministic dynamics

With $\upsilon_q = 0$, the economy is deterministic in $(h, q)$ and evolves according to ODEs (paper's eqs. 19–20). The qualitative implication is: experts (higher $\rho_e$) consume faster, so $h_t$ drifts downward, but prices adjust smoothly and there are no crises because capital always stays with experts.

## 5 Brownian Sunspot Equilibrium (BSE) (Section 2)

### 5.1 Step 1: Markov pricing $q = q(h)$ implies the key consistency condition

Conjecture $q_t = q(h_t)$. By Itô,

$$\sigma_q = \frac{q'(h)}{q(h)}\,\sigma_h, \qquad \upsilon_q = \frac{q'(h)}{q(h)}\,\upsilon_h.$$

Since $\sigma_h \equiv 0$ by (12), we must have

$$\sigma_q \equiv 0. \tag{14}$$

Combining $\upsilon_h = (\kappa - h)\upsilon_q$ with $\upsilon_q = (q'/q)\upsilon_h$ yields the paper's critical condition:

$$\left[1 - (\kappa - h)\frac{q'(h)}{q(h)}\right]\upsilon_q(h) = 0. \tag{15}$$

Thus either:

(i) $\upsilon_q(h) = 0$ (no sunspot risk locally), or

(ii) $1 = (\kappa - h)\, q'(h)/q(h)$ (sunspot volatility can be nonzero).

### 5.2 Step 2: where can sunspot volatility exist?

- If $\kappa = 1$ (experts manage all capital), then (PO) implies $q(h) = a_e/\bar{\rho}(h)$, which is decreasing in $h$. But $1 = (1-h)q'/q$ would require $q'(h) > 0$, a contradiction. Hence $\upsilon_q = 0$ whenever $\kappa = 1$.

- If $\kappa < 1$ (households hold some capital), then $\upsilon_q = 0$ is impossible because (RB) would force $\kappa = 1$. So $\upsilon_q \neq 0$ and the second branch of (15) must hold.

Therefore, in a BSE:

$$\upsilon_q(h) \neq 0 \text{ only in the inefficient “fire-sale” region where } \kappa(h) < 1.$$

### 5.3 Step 3: ODE for the pricing function in the fire-sale region

In the region $\kappa < 1$, use the price–output relation (PO) to write

$$\kappa(h) = \frac{q\bar{\rho}(h) - a_h}{a_e - a_h}.$$

Impose the nontrivial branch of (15), $1 = (\kappa - h)q'/q$, and substitute $\kappa(h)$ to obtain the paper's first-order ODE:

$$q'(h) = \frac{(a_e - a_h)\, q(h)}{q(h)\bar{\rho}(h) - ha_e - (1-h)a_h}, \qquad \text{for } \kappa(h) < 1. \tag{16}$$

**Boundary condition = “disaster belief.”** The paper parametrizes equilibria by a boundary condition for $\kappa(h)$ as $h \to 0$:

$$\lim_{h \downarrow 0} \kappa(h) = \kappa_0 \in [0, 1).$$

This $\kappa_0$ is interpreted as a belief about what happens in the “tail scenario” where experts are severely undercapitalized. The baseline BSE uses $\kappa_0 = 0$ (experts fully delever in the limit).

### 5.4 Step 4: pinning down the *level* of volatility via risk-balance

When $\kappa < 1$, the risk-balance condition (RB) binds and yields a closed-form expression for sunspot volatility:

$$\upsilon_q^2(h) = h(1-h)\,\frac{\kappa(h) - h}{q(h)}\,(a_e - a_h), \qquad \text{for } \kappa(h) < 1. \tag{17}$$

*Interpretation:* the productivity gap $(a_e - a_h)$ makes misallocation costly and therefore makes prices highly sensitive to reallocations; this feeds into larger endogenous volatility.

### 5.5 Existence/uniqueness and stationarity (Proposition 1)

- The ODE (16) has a singularity at $h \downarrow 0$ under $\kappa_0 = 0$, so the paper proves existence/uniqueness by approximation (Appendix A).
- The solution features an endogenous cutoff $h^\star$ such that:

$$\kappa(h) < 1,\ \upsilon_q(h) > 0 \text{ for } h \in (0, h^\star), \qquad \kappa(h) = 1,\ \upsilon_q(h) = 0 \text{ for } h \in [h^\star, 1).$$

- The state process $h_t$ has a non-degenerate stationary distribution on $(0, h^\star]$ because experts receive a risk premium when undercapitalized but also discount faster.

### 5.6 Interpretation as self-fulfilling fire sales (Section 2.2)

The paper's feedback loop can be summarized as:

$$\kappa \downarrow \Rightarrow q \downarrow \Rightarrow h \downarrow \Rightarrow \kappa \downarrow,$$

where:

- $\kappa \downarrow \Rightarrow q \downarrow$ comes from the price–output relation (PO).
- $q \downarrow \Rightarrow h \downarrow$ comes from non-hedgeable exposure to $\mathrm{d}Z_t$: experts are the levered holders of capital.
- $h \downarrow \Rightarrow \kappa \downarrow$ is the belief/coordination component (disaster belief): agents think experts sell more when undercapitalized.

### 5.7 Disaster-belief multiplicity (Section 2.3)

For each $\kappa_0 \in [0, 1)$ there is a unique BSE($\kappa_0$) (Corollary 1). As $\kappa_0 \to 1$ the equilibrium converges to the safe FE; as $\kappa_0 \to 0$ it converges to the baseline BSE with the largest fire-sale region.

## 6 Link to conventional accelerator equilibria (Section 3)

### 6.1 Conventional accelerator equilibrium (CAE)

A CAE shuts down hedging: $x_{e,t} = x_{h,t} = 0$. Then fundamental volatility is no longer shared, and it enters equilibrium through the total return volatility $\sigma + \sigma_q$.

In a Markov equilibrium $q = q(h)$, the paper derives the amplification identity:

$$\sigma_q = \frac{(\kappa - h)\, q'(h)/q(h)}{1 - (\kappa - h)\, q'(h)/q(h)}\, \sigma. \tag{18}$$

*Interpretation:* this takes the form of a geometric series: initial wealth losses move $h$, which moves $q$, which further moves $h$, etc.

### 6.2 Observational equivalence as fundamental risk vanishes (Proposition 2)

As $\sigma \to 0$ in a CAE, (18) forces either $\sigma_q \to 0$ (safe) or

$$1 - (\kappa - h)\, q'(h)/q(h) \to 0,$$

which is exactly the nontrivial sunspot branch that characterizes the BSE. Under the baseline boundary condition $\kappa_0 = 0$, the CAE converges in distribution to the BSE(0) as $\sigma \to 0$.

**Take-away.** Even with complete markets for fundamental risk, "accelerator-looking" dynamics survive as self-fulfilling equilibria. The conventional financial accelerator can be interpreted as selecting a particular sunspot equilibrium in the small-risk limit.

# 7 Refinement and equilibrium selection (Section 3.2)

## 7.1 Perceived accelerator equilibrium (PAE)

Introduce a vanishingly small *perceived* fundamental impact of the sunspot shock:

$$dk_{i,t} = k_{i,t}\big(g \; dt + \sigma \; dW_t + V \; dZ_t\big), \tag{19}$$

where in truth $V = 0$, but agents perceive $V > 0$ and cannot hedge $dZ_t$. This turns $Z_t$ into an (unspanned) fundamental risk *in their beliefs*.

## 7.2 Selection logic

(i) For each $V > 0$, agents effectively behave as in a conventional accelerator model with an unspanned risk factor.

(ii) Imposing a mild leverage/limited-commitment refinement (developed in Appendix C.3) selects the "maximal fire-sale" boundary condition $\kappa_0 = 0$.

(iii) As $V \to 0$ and beliefs converge to rational expectations, the only limiting equilibrium is the baseline BSE(0) (Proposition 3).

**Interpretation.** Even an arbitrarily small perceived unspanned risk can eliminate the safe equilibrium, forcing coordination on a crisis equilibrium. The crisis then remains even as the perceived risk vanishes.

# 8 "Large" fire sales with Poisson jumps (Section 4)

## 8.1 Jump specification

Replace Brownian shocks with a Poisson process $J_t$ of intensity $\lambda$. Capital evolves as

$$\frac{dK_t}{K_{t^-}} = g \; dt - z \; dJ_t,$$

with $z \geq 0$. If $z = 0$, the jump is a *sunspot*.

Capital prices jump:

$$\frac{dq_t}{q_{t^-}} = m_q(h)\, dt - z_q(h)\, dJ_t,$$

and the wealth share jumps as

$$\frac{dh_t}{1} = m_h(h)\, dt - z_h(h)\, dJ_t, \qquad z_h(h) = h_{t^-} - h_t.$$

Define the negative jump in the return on (value of) capital:

$$z_R := z + z_q - z z_q. \tag{20}$$

### 8.2 Jump risk-balance and fixed point

Portfolio optimality yields a jump analogue of risk balance:

$$0 = \min \left\{ 1-\kappa,\ \frac{a_e - a_h}{q} - \frac{\kappa - h}{h(1-h)} \underbrace{\frac{\lambda z_R^2}{\left(1 - \frac{1-\kappa}{h} z_R\right)\left(1 - \frac{1-(1-\kappa)}{1-h} z_R\right)}}_{\text{effective unspanned jump risk}} \right\}. \tag{RBJ}$$

The wealth-share jump satisfies

$$z_h = \frac{(\kappa - h) z_R}{1 - z_R}, \tag{21}$$

and Markov pricing implies the price jump is pinned down by the post-jump state $\hat{h} = h - z_h$:

$$z_q = \frac{q(\hat{h}) - q(h)}{q(h)}. \tag{22}$$

Equations (RBJ)–(22) form a fixed point capturing the two-way feedback between fire sales and prices in jumps.

### 8.3 Poisson sunspot equilibria (PSE) and limit as $z \to 0$

A **PSE** is an equilibrium with $z = 0$ (no fundamental jump) but $z_q(\cdot) \not\equiv 0$ (sunspot jump in prices). Numerically, the paper shows that as the fundamental jump size $z \to 0$, the CAE converges to a PSE, paralleling Proposition 2.

### 8.4 Why misallocation can be worse under jumps

Jumps generate left-skewed losses: capital and prices can fall sharply in a single event. Even if the economy is "healthy" today (high $h$), the possibility of a large discrete event tomorrow depresses prices and expert capital share.

## 9 Appendix models and robustness: what is added beyond the baseline

### 9.1 Appendix C.1: "Hedging" equilibria with $\sigma_q < 0$

In the CAE (no hedging), only the variance $(\sigma + \sigma_q)^2$ matters in risk balance, so the sign of $(\sigma + \sigma_q)$ need not be positive. Appendix C.1 numerically constructs an equilibrium where $\sigma_q < 0$ in the fire-sale region, so prices move opposite to the fundamental shock. This corresponds to a modified ODE for $q'(h)$ (paper's eq. C.4).

## 9.2 Appendix C.2: disaster-belief indeterminacy in CAEs

Just as BSEs form a continuum indexed by $\kappa_0$, conventional equilibria (CAEs) also admit multiple "disaster beliefs". Different $\kappa_0$ deliver different fire-sale magnitudes and different stationary distributions.

## 9.3 Appendix C.3: limited commitment and a leverage constraint as a refinement

A limited-commitment / diversion friction implies a leverage cap

$$\frac{q_t k_{i,t}}{n_{i,t}} \leq \beta.$$

The appendix shows that letting $\beta \to \infty$ (a vanishing friction) *selects* the maximal fire-sale boundary $\kappa_0 = 0$ via an existence/nonexistence argument: for $\kappa_0 > 0$ and $\beta$ large enough, no equilibrium can satisfy both the constraint and the candidate pricing function.

## 9.4 Appendix D: CRRA preferences (risk aversion $\gamma \neq 1$)

The paper extends existence and numerical properties to CRRA utility, showing sunspot equilibria persist when agents are not log. Risk aversion $\gamma$ and disaster belief $\kappa_0$ shift: (i) the size of the fire-sale region, (ii) the level of prices, (iii) endogenous volatility, and (iv) the stationary distribution of $h$.

## 9.5 Appendix F: partial equity issuance / "approaching complete markets"

Experts can offload a fraction of their risk by issuing equity; a parameter $\chi$ measures the remaining friction. Two key messages:

(i) For any $\chi > 0$ there remains a nontrivial fire-sale region (endogenous amplification persists).

(ii) As $\chi \to 0$, the probability of visiting the fire-sale region can shrink (crises become rarer), even if the region does not literally vanish.

# 10 Interpretation of all figures

## 10.1 Figure 1: BSE vs FE (pricing, volatility, stationary distribution)

Figure 1 has three panels:

(a) $q(h)$: In the FE (green dashed), $q(h) = a_e/\bar{\rho}(h)$ is decreasing in $h$ (more expert wealth implies higher discounting and lower price). In the BSE (blue), $q(h)$ is *increasing* for low $h$ because a fall in expert wealth triggers selling and misallocation to households, lowering output efficiency and thereby lowering prices. At the cutoff $h^\star$, the BSE coincides with the FE line because $\kappa(h) = 1$ and volatility shuts down.

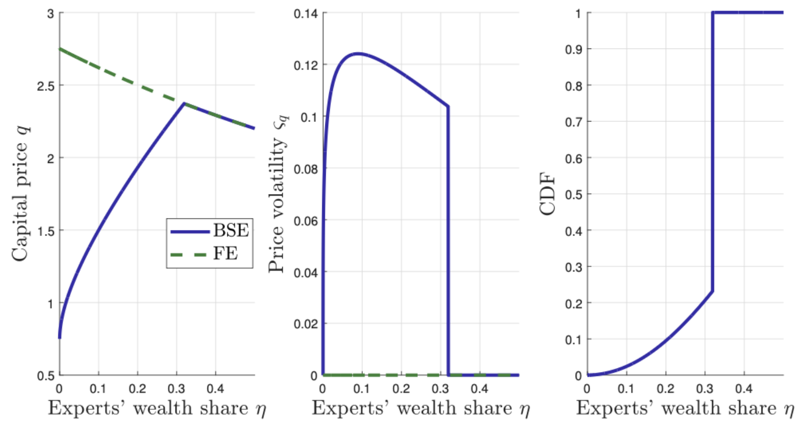


Figure 1: Capital price $q$, volatility $\varsigma_q$, and stationary CDF of $\eta$ in the Brownian Sunspot Equilibrium (BSE) and Fundamental Equilibrium (FE). The stationary CDF is calculated using the Kolmogorov Forward Equation for $\eta$. Parameters: $\rho_e = 0.06$, $\rho_h = 0.04$, $a_e = 0.11$, $a_h = 0.03$. (Note that $g$ and $\sigma$ are irrelevant to the solution.)

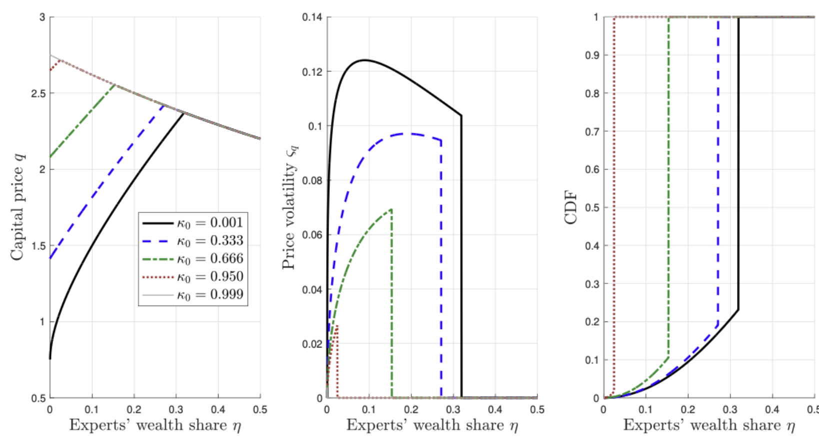


Figure 2: Capital price $q$, sunspot volatility $\varsigma_q$, and stationary CDFs of $\eta$ for different levels of disaster belief $\kappa_0$. Parameters: $\rho_e = 0.06$, $\rho_h = 0.04$, $a_e = 0.11$, $a_h = 0.03$.

(b) $v_q(h)$: Sunspot volatility is strictly positive only in the fire-sale region $h < h^\star$ and exactly zero when $\kappa = 1$.

(c) Stationary CDF of $h$: The state spends time in the undercapitalized region but does not collapse to $h = 0$ because experts earn a risk premium when they are scarce.

### 10.2 Figure 2: the continuum of BSEs indexed by disaster belief $\kappa_0$

Figure 2 overlays several BSE($\kappa_0$) solutions. As $\kappa_0$ rises toward 1, the fire-sale region shrinks, volatility drops, and the stationary distribution concentrates at higher $h$ (closer to the safe FE). As $\kappa_0$ approaches 0, the equilibrium approaches the baseline BSE with the largest endogenous volatility.

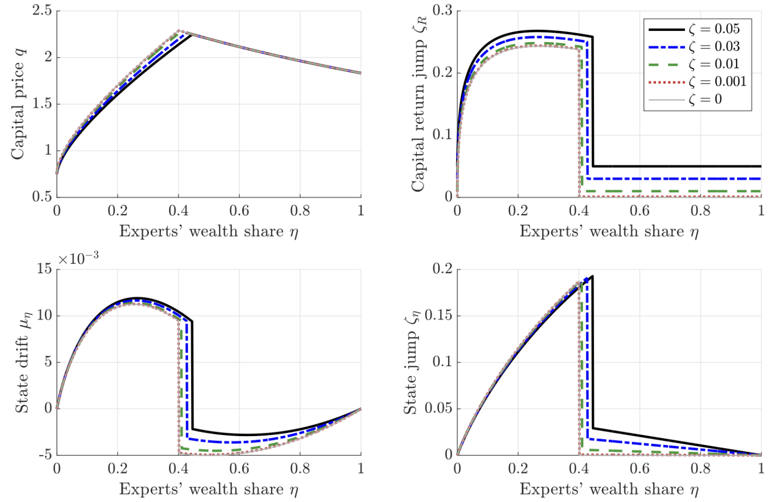


Figure 3: Convergence to a Poisson Sunspot Equilibrium as $\zeta \to 0$. Parameters: $\rho_e = 0.06$, $\rho_h = 0.04$, $a_e = 0.11$, $a_h = 0.03$, $\lambda = 0.1$. In all cases, we use the boundary condition $\kappa(0) = 0$.

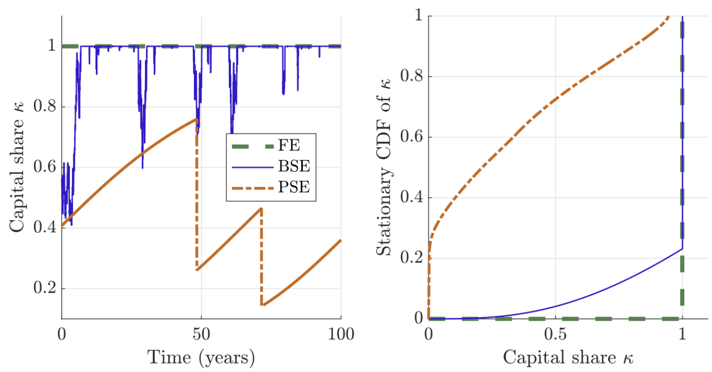


Figure 4: Time series and stationary density of capital price $q$ in a PSE, BSE, and Fundamental Equilibrium (FE). Parameters: $\rho_e = 0.06$, $\rho_h = 0.04$, $a_e = 0.11$, $a_h = 0.03$, $\sigma = 0$, $\zeta = 0$. We use the boundary condition $\kappa(0) = 0$ in all cases. For the PSE, we use $\lambda = 0.02$ as the arrival rate of Poisson jumps and compute the stationary CDF via a 100,000 year simulation at the weekly frequency.

## 10.3 Figure 3: convergence to a Poisson sunspot equilibrium as $z \to 0$

Figure 3 studies the jump model for various fundamental jump sizes $z$:

- The pricing function $q(h)$ converges to a non-FE limit as $z \to 0$.
- Even when $z$ is tiny, the implied equilibrium return jump $z_R(h)$ can remain nontrivial in the fire-sale region, reflecting endogenous jump risk.
- The drift and jump size of the state ($\mu_h$ and $z_h$) illustrate the "run-like" movement in expert wealth when crises occur.

## 10.4 Figure 4: capital misallocation under PSE vs BSE vs FE

Despite a caption typo referring to $q$, the plotted object is the *expert capital share* $\kappa$. The left panel simulates $\kappa_t$ over time; the right panel shows its stationary distribution/CDF.

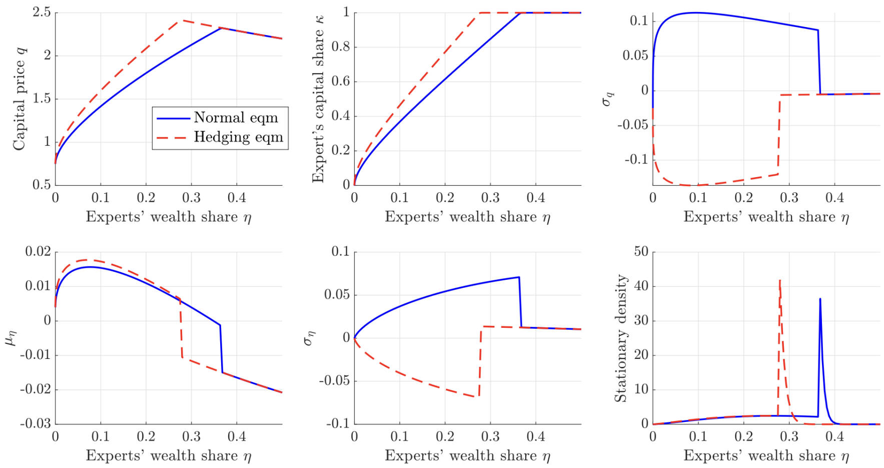


Figure C.1: Two equilibria (normal versus hedging) both with boundary condition $\kappa(0) = 0$. Parameters: $\rho_e = 0.06$, $\rho_h = 0.04$, $a_e = 0.11$, $a_h = 0.03$, $\sigma = 0.025$.

- FE: $\kappa \equiv 1$ (efficient allocation).
- BSE: mostly $\kappa = 1$ but with occasional dips in the fire-sale region.
- PSE: deeper and more persistent drops in $\kappa$ because jump risk is left-skewed (large discrete crises), worsening misallocation.

### 10.5 Figure C.1: normal vs hedging CAE equilibria

Figure C.1 compares two CAEs with the same boundary condition $\kappa(0) = 0$:

- *Normal equilibrium:* $\sigma_q > 0$ in the fire-sale region; prices co-move with the fundamental shock.
- *Hedging equilibrium:* $\sigma_q < 0$ in the fire-sale region; prices move opposite to the fundamental shock.

The remaining panels show how this sign change affects expert capital share, state drift/volatility, and the stationary density of $h$.

### 10.6 Figure C.2: CAEs with different disaster beliefs

Analogous to Figure 2 but for CAEs: varying $\kappa_0$ changes the fire-sale region, price sensitivity, and the stationary distribution. Larger $\kappa_0$ makes the economy closer to efficient allocation ($\kappa$ near 1) and reduces endogenous volatility.

### 10.7 Figure C.3: refinement via limited commitment / leverage constraint

Figure C.3 illustrates why a leverage constraint can *eliminate* equilibria with $\kappa_0 > 0$ for large enough $\beta$. The candidate constrained pricing function (blue) must lie below the leverage-implied upper bound in the undercapitalized region. If it does not, no equilibrium exists; the only robust surviving boundary in the $\beta \to \infty$ limit is $\kappa_0 = 0$.

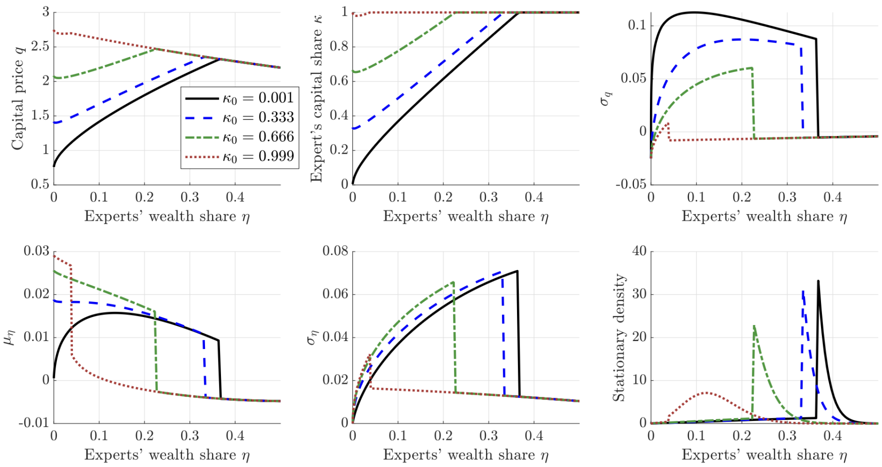


Figure C.2: Conventional equilibria with different disaster beliefs $\kappa_0$. Parameters: $\rho_e = 0.06$, $\rho_h = 0.04$, $a_e = 0.11$, $a_h = 0.03$, $\sigma = 0.025$.

### 10.8 Figures D.1 and D.2: CRRA robustness (risk aversion and disaster beliefs)

These figures show that the qualitative structure (fire-sale region + zero-volatility efficient region) persists beyond log utility.

- Higher risk aversion $\gamma$ tends to shrink experts' effective risk-bearing capacity, shifting the cutoff and changing the magnitude of endogenous volatility and prices (Figure D.1).
- Varying disaster beliefs $\kappa_0$ produces the same multiplicity logic as in log utility: more optimistic tail beliefs imply smaller crises (Figure D.2).

### 10.9 Figure F.1: approaching complete markets via equity issuance

Figure F.1 studies equilibria as the equity-issuance friction $\chi \to 0$. The key visual message is that pricing and leverage objects converge (in scaled coordinates) to nontrivial limits, but the stationary distribution shifts so that the economy may spend less probability mass in the crisis region as markets become closer to complete.

## 11 Takeaway

Even when all *fundamental* risks can be hedged, financial accelerator dynamics can survive because markets cannot hedge *endogenously coordinated* (sunspot) shocks. In the model, a sunspot-driven price drop hits experts disproportionately, forcing them to sell capital to less productive households, which further lowers prices and validates the original fear. This yields a Brownian sunspot equilibrium (and a Poisson jump analogue) with fire sales and misallocation, alongside a safe fundamental equilibrium. A small-noise refinement—agents slightly misperceive sunspot risk as fundamental and face a vanishingly mild leverage friction—selects the maximal fire-sale equilibrium, showing that "crisis" outcomes can be the unique limit even as perceived fundamental risk vanishes.

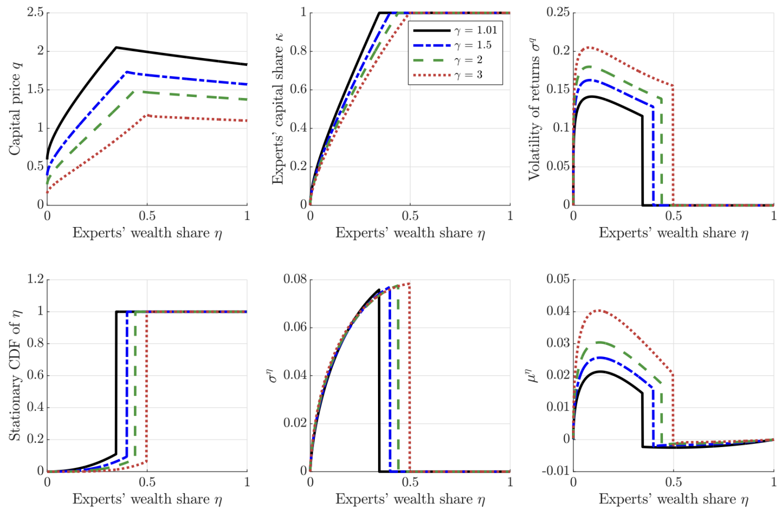


Figure D.1: Sunspot equilibrium for different risk aversion $\gamma$. The disaster belief is set to $\kappa_0 = 0.001$. Other parameters: $a_e = 0.11$, $a_h = 0.03$, $\rho_e = 0.06$, $\rho_h = 0.05$, $g = 0.02$.

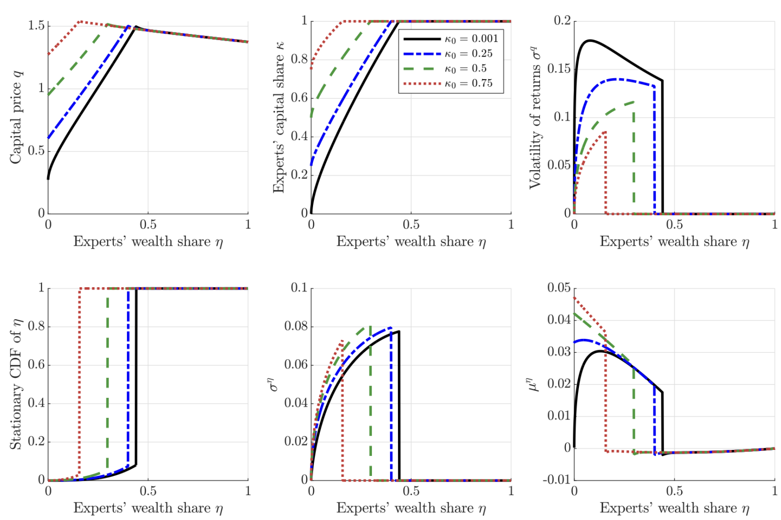


Figure D.2: Sunspot equilibrium for different disaster beliefs $\kappa_0$. Risk aversion is set to $\gamma = 2$. Other parameters: $a_e = 0.11$, $a_h = 0.03$, $\rho_e = 0.06$, $\rho_h = 0.05$, $g = 0.02$.

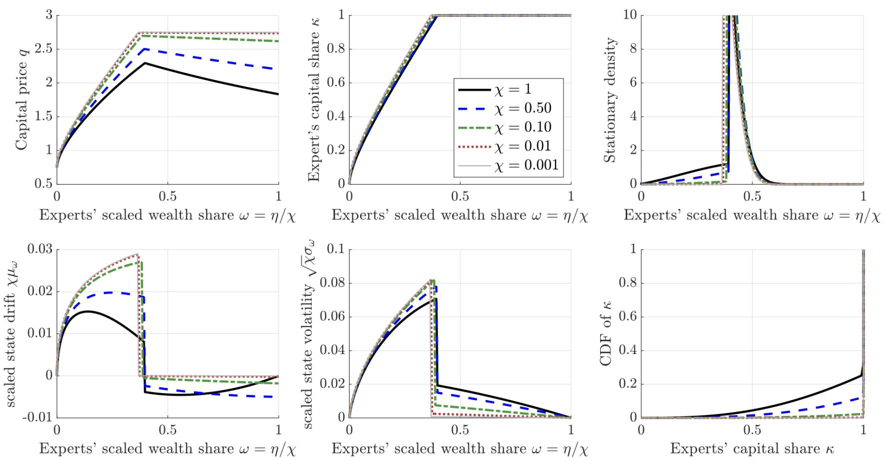


Figure F.1: Equilibria as equity-issuance frictions improve, $\chi \to 0$. Parameters: $\rho_e = 0.06$, $\rho_h = 0.04$, $a_e = 0.11$, $a_h = 0.03$, $\sigma = 0.04$. In all cases, we use the boundary condition $\kappa(0) = 0$.

# 12 What the paper concludes

The paper delivers a sharp message about the origin of fire sales in macro–finance models.

(i) **If all *fundamental* aggregate risks are hedgeable, fundamentals cannot generate fire sales.** In the baseline environment, agents can frictionlessly hedge the fundamental capital-quality shock. As a result, the conventional financial-accelerator logic—where fundamentals move intermediaries' balance sheets and trigger forced sales—breaks down.

(ii) **Even so, fire sales can still arise as *self-fulfilling* equilibria.** The model admits a *sunspot equilibrium* in which agents coordinate on non-fundamental shocks that are not hedgeable. These shocks move asset prices, shift wealth between experts and households, and induce experts to sell capital to less productive households, generating misallocation and endogenous volatility.

(iii) **Equilibrium multiplicity is structured by "disaster beliefs."** There is a continuum of "partial fire-sale" equilibria indexed by a boundary condition—a belief about what happens when experts become extremely undercapitalized. The benchmark equilibrium corresponds to the *worst-case* tail scenario (a "full" fire sale).

(iv) **A simple refinement selects the fire-sale equilibrium (and eliminates the safe one).** Although the model features a safe, efficient equilibrium with no fire sales, a trembling-hand style refinement (vanishingly small limited-commitment / leverage friction together with agents perceiving the sunspot to have a vanishingly small fundamental component) selects the worst fire-sale equilibrium as the unique limiting outcome, ruling out the safe equilibrium and all partial fire-sale equilibria.

(v) **The resulting dynamics resemble runs, but the policy logic differs.** The equilibrium looks run-like in that households "withdraw" funding (by reallocating toward capital) at the same time that experts are forced to sell. However, the mechanism does not require maturity mismatch or default, so classic deposit insurance logic is not the natural fix.

## 13 Economic intuition: why self-fulfilling fire sales survive complete hedging of fundamentals

**Step 1: separate hedgeable and unhedgeable uncertainty.** The environment contains (at least) two conceptually distinct shocks:

- a **fundamental shock** to capital quality that *can be hedged* via a spanning market;
- a **sunspot shock** that is *not directly hedgeable* because no market exists for it.

Once fundamental risk is spanned, it cannot move the *distribution of wealth* across agents (experts vs households). So the conventional accelerator route "fundamentals ⇒ wealth redistribution ⇒ forced sales" is shut down.

**Step 2: sunspots matter because they create an *unspanned* balance-sheet risk.** Experts are the high-productivity owners/managers of capital, so their balance sheets are disproportionately exposed to the asset price of capital. If agents believe that a sunspot can move prices, then it becomes an unhedgeable source of risk *precisely because it is not traded.*

**Step 3: the key feedback loop is balance-sheet mediated and self-justifying.** The self-fulfilling mechanism is:

$$\kappa \downarrow \Rightarrow \ q \downarrow \Rightarrow \ \eta \downarrow \Rightarrow \ \kappa \downarrow .$$

Here:

- $\kappa$ is the experts' share of capital (misallocation rises when $\kappa$ falls),
- $q$ is the capital price (lower when households hold more capital, because households are less productive),
- $\eta$ is experts' wealth share (falls when capital prices fall, because experts are effectively leveraged in capital).

If a sunspot triggers (or is expected to trigger) selling by others, the resulting price decline reduces experts' wealth, tightening their effective constraint and rationalizing their own selling. In equilibrium, the *belief* that a non-fundamental shock will cause a fire sale is enough to make it happen.

**Step 4: why the mechanism is intrinsically dynamic.** The equilibrium requires future uncertainty in asset prices. If the horizon were finite and the terminal capital price were pinned down, backward induction would make earlier prices effectively riskless, and the economy would collapse to the safe equilibrium. Thus the fire-sale equilibrium is inherently about intertemporal risk, balance sheets, and forward-looking pricing.

## 14 How the paper reframes the financial accelerator

The paper overturns a common inference:

> "If aggregate risks are hedgeable, the accelerator disappears, so fire sales must be fundamentally driven by unspanned aggregate risk."

Instead, the paper argues:

> *Fire sales can be inherently self-fulfilling: even if all fundamental risks are spanned, any non-hedgeable emergent shock can sustain accelerator-like fluctuations.*

In this sense, the traditional assumption that "fundamental shocks are unhedgeable" may not be the deep driver of crisis dynamics; it can be a modeling shortcut that indirectly captures the consequences of unspanned or emergent risks.

## 15 Policy intuition: why "deposit insurance" is not the right fix here

The run-like dynamics in the model are not driven by depositors coordinating on default of short-term debt. Debt is repaid, so insuring deposits does not address the central coordination failure. Instead, the model points to policy tools that change *disaster beliefs* (the tail boundary condition governing how severe the fire sale is expected to be). A natural candidate is a **state-contingent asset-market backstop** (e.g., credible promises to purchase assets in disasters), which can shrink the set of self-fulfilling fire-sale beliefs by putting a floor under the tail price of capital. A key open question is the moral-hazard tradeoff: whether such promises reduce crises ex post but worsen risk-taking ex ante.

## 16 Summary

*Even with perfect hedging of all fundamental aggregate shocks, fire sales can arise and persist as a dynamically self-fulfilling response to unhedgeable non-fundamental shocks, and a simple trembling-hand refinement selects the worst fire-sale equilibrium as the unique limiting outcome.*